



MANIPAL SCHOOL OF INFORMATION SCIENCES

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Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems)

Course Plan

Course Name	:	Real Time Operating Systems
Course Code	:	ESE 5104
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Mr. Shreyas N
Name of the Program Coordinator	:	Dr. B Dinesh Rao

Date:	1.8.2026
Signature of Program Coordinator	
with Date	

Date:	30.07.2026
Signature of Course Coordinator	
with Date	



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Program Education Objectives (PEOs)



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The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems), program are as follows

PEO No.	Education Objective
PEO 1	Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in embedded systems.
PEO 2	Introduce state of art technologies in the area of embedded systems and inculcate ethical practices to make industry-ready professionals.
PEO 3	Promote scientific and societal advancement through research and entrepreneurship.

Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems), graduates will be able to:



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PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop and implement embedded systems requirements based on theoretical principles and practical knowledge.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analyzing and decision-making.

1. Course Plan

1.1 Primary Information

Course Name	:	Real Time Operating Systems [ESE 5104]
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L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	C programming
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No.ofContact Hours	Program Outcomes (POs)	BL
CO1	Evaluate the features of OS, RTOS, processes and process scheduling algorithms.	15	P03	5
CO2	Analyse various techniques for addressing synchronization, deadlock and memory management.	16	P04	4
CO3	Evaluate schedulability in real time systems.	5	P05	5

1.3 Assessment Plan

Components	Mid semester exam (MSE)	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	2 weeks	180 minutes



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Weightage	0.3	0.2	0.5
Typology of questions	Analyse, evaluate	Analyse, evaluate	Analyse, evaluate
Pattern	Answer all 5 questions	Two weeks	Answer all 10 questions
Schedule	As per academic calendar.	Assignment submission: september 30th, october 25th	As per academic calendar.
Topics covered	Introduction to OS and RTOS, Process Management, Multithreaded Programming, RTOS	Introduction to OS and RTOS, Process Management, Multithreaded Programming, Synchronization:, memory management, deadlocks and Real Time Systems	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.

1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Introduction to OS and RTOS - Essential features of an OS	CO1



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L2	Single Processor Systems and Multiprocessor Systems	C01
L3	Essential Features of Batch Processing, Time sharing, Multiprogramming Interactive systems	C01
L4	User mode and Kernel Mode operations,	C01
L5	Distinction between function call and system call	C01
L6	Real time operating system and real time embedded systems	C01
L7	Process Management: A process in memory, process state	C01
L8	PCB, Process scheduling, scheduling Queues	C01
L9	Types of schedulers, Process system calls	C01
L10	IPC using Shared Memory, IPC using Sockets.	C01
L11	Process Scheduling: Introduction, scheduling criteria	C01
L12	scheduling Algorithms – FCFS, SJF, PS, RR,	C01
L13	scheduling Algorithms – FCFS, SJF, PS, RR,	C01
L14	Multilevel Queues, Multilevel feedback Queue Scheduling, Scheduling evaluations.	C01
L15	Multilevel Queues, Multilevel feedback Queue Scheduling, Scheduling evaluations.	C01
L16	Multithreaded Programming: Introduction, benefits,	C02



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L17	multithreading models, Pthreads, Win32 threads,	C02
L18	Threading Issues, Thread pools Linux threads.	C02
L19	Synchronization: Introduction, Critical Section Problem	C02
L20	Petersons Solutions, synchronization hardware	C02
MSE	Mid Semester Exam	CO 1, CO 3
L21	Semaphores, usage, implementations	C02
L22	Deadlocks and starvation,	C02
L23	Classical problem of synchronization – Bounded Buffer problem, Reader's Writer's problem, Dining Philosophers problem, sleeping barbers problem; Monitors.	C02
L24	Classical problem of synchronization – Bounded Buffer problem, Reader's Writer's problem, Dining Philosophers problem, sleeping barbers problem; Monitors.	C02
L25	Classical problem of synchronization – Bounded Buffer problem, Reader's Writer's problem, Dining Philosophers problem, sleeping barbers problem; Monitors.	C02
L26	Deadlocks: Introduction, deadlock, characterization	C02
L27	methods for handling deadlocks, deadlock prevention,	C02
L28	deadlock avoidance, recovery from deadlock	C02
L29	Memory Management: Memory Management Strategies, Virtual Memory Management	C02
L30	Memory Management: Memory Management Strategies, Virtual Memory Management	C02



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L31	Memory Management: Memory Management Strategies, Virtual Memory Management	C02
L32	Real Time Systems: Overview of Real Time Systems	C03
L33	Real Time clocks and Real Time Scheduling Algorithms – RMS, DMS, EDF	C03
L34	Real Time clocks and Real Time Scheduling Algorithms – RMS, DMS, EDF	C03
L35	Real Time clocks and Real Time Scheduling Algorithms – RMS, DMS, EDF	C03
L36	priority inheritance	C03

1.5 References



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1. A. Silberschatz, P. Galvin, and G. Gagne, Operating System Principles, 7th ed., John Wiley & Sons, 10th Edition, 2018.
2. A. Burns and A. Wellings, Real-Time Systems and Programming Languages, 4th ed., Addison-Wesley, 2009.
3. M. Milenkovic, Operating Systems: Concepts and Design, McGraw-Hill, 1992.
4. M. J. Bach, The Design of the UNIX Operating System, Prentice-Hall, 1986.
5. B. W. Kernighan and D. M. Ritchie, The C Programming Language, 2nd ed., Prentice-Hall, 1988.
6. MOOC:<https://in.coursera.org/learn/real-time-embedded-systems-concepts-practices#syllabus>

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.

1.7 Course Timetable



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I Semester						Room: LG01-LH08			
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday									
Tuesday									
Wednesday									
Thursday									
Friday									
Saturday									

RTOS 1.5hr

RTOS 1.5hr

1.8 Assessment Plan

COs	Marks & Weightage
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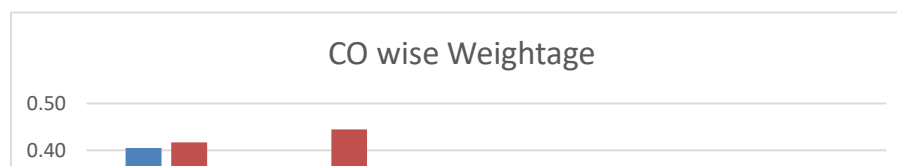


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CO No.	CO Name	MSE Marks (Max. 50)	Assignment Marks (Max. 20)	ESE Marks (Max. 100)	Actual CO wise Weightage	Expected CO wise Weightage
CO1	Evaluate the features of OS, RTOS, processes and process scheduling algorithms.	30	5	35	0.41	0.42
CO2	Analyse various techniques for addressing synchronization, deadlock and memory management.		10	40	0.30	0.44
CO3	Evaluate schedulability in real time systems.	20	5	25	0.30	0.14
	Marks (weightage)	0.3	0.2	0.5	1.00	1.00

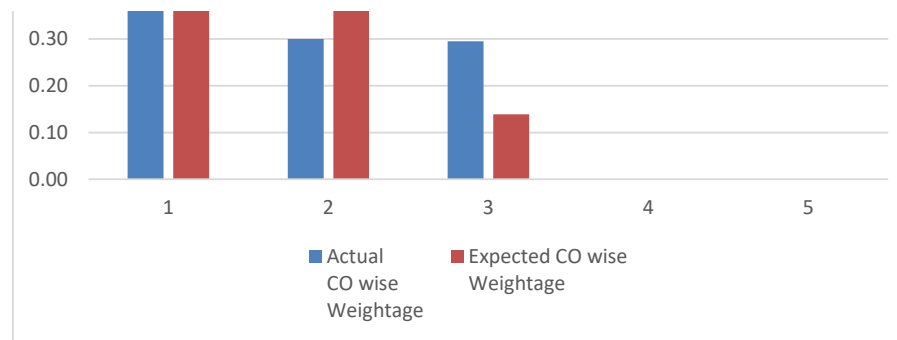




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1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

SI. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	MSE	0.3	1	• Performance is measured using MSE attainment level. • Reference: question paper and answer scheme. • MSE is assessed for a maximum of 50 marks and scaled down to 30 marks.



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2	Assignments	0.2	2	• Performance is measured using assignments/quiz attainment level. • Assignments/quiz are evaluated for a maximum of 20 marks.
3	ESE	0.5	1	• Performance is measured using ESE attainment level. • Reference: question paper and answer scheme. • ESE is assessed for a maximum of 100 marks and scaled down to 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1			Y		
CO2				Y	
CO3					Y
Average Articulation Level			*	*	*